

Curriculum Vitae

1. PERSONAL INFORMATION:

Dr. Roshni Rathore

Ward number 11 Pandalpur Rahil, Sagar M.P. 470227

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DOB: 20/09/1995

Gender: Female



https://scholar.google.com/citations?hl=en&view_op=search_authors&mauthors=label:synthesis_and_characterization_of_nanomaterials_for_various_applications

2. ACADEMIC QUALIFICATION:

Degree	Year	Percentage	University/Institute	Field of Specialization
10 th	2011	68%	M. P. Board	Hindi, English, Math, Science, SST, Sanskrit
12 th	2013	78.4%	M. P. Board	Bio-Group
B. Sc.	2016	70.1%	Dr. Harisingh Gour Vishwavidyalaya Sagar, (M. P.)	Chemistry, Botany, Zoology
M. Sc.	2018	61.95%	Dr. Harisingh Gour Vishwavidyalaya Sagar, (M. P.)	Chemistry
Ph. D.	2019-24	72.25	Dr. Harisingh Gour Vishwavidyalaya Sagar, (M. P.)	Thesis Title: “Synthesis and Characterization of Manganese Oxides Based Nanocomposites for Various Applications”

3. AWARD:

Madhya Pradesh State Eligibility Test (MPSET) -2023.

4. AREA OF SPECIALIZATION:

Synthesis and Characterization of nanocomposites for Wastewater treatment by Adsorption and Photocatalytic Degradation method.

5. LABORATORY SKILLS:

Synthesis of nanoparticles and nanocomposites performed UV-VIS Spectroscopy, and various parameters like kinetics, adsorption isotherms, thermodynamics, and so on for adsorption and degradation experiments.

6. Research Experience:

- Having four years of research experience in chemistry from 30/09/2019 to 30/09/2023 at the Department of Chemistry, Dr. Harisingh Gour Central University, Sagar M.P.

7. Teaching Experience:

- Having one year of teaching experience from 01/09/2024 to 05/09/2025 at Govt. Polytechnic college Dewas, M.P.
- Having half year of teaching experience from 22/02/2024 to 22/08/2024 at **Oriental group of Institute Bhopal, M.P.**

8. PUBLICATIONS

- I. **R. Rathore**, A. Waghmare, S. Rai, V. Chandra, Removal of nitrobenzene from aqueous solution using manganese ferrite nanoparticles, **Inorg. Chem. Commun.** 153 (2023) 110848 (**IF: 5.4**).
- II. **R. Rathore**, A. Waghmare, S. Rai, and V. Chandra, "Synthesis of α -MnO₂, β -C₂S, and α MnO₂/ β -C₂S nanocomposite for effective removal of phosphate from water," **Int. J. Appl. Ceram. Technol.** 21 (2023) 142-157 (**IF: 2.32**).
- III. **R. Rathore**, A. Waghmare, S. Rai, V. Chandra, "Enhanced Photocatalytic Performance of gC₃N₄/MnFe₂O₄ Nanocomposite for Crystal Violet Dye Degradation under Solar Light" **Chemistry Select.** 8 (2023) e202301344 (**IF: 2.30**).
- IV. A. Waghmare, **R. Rathore**, A. Pandey, V. Chandra, Graphene-Oxide-Coated, PolypyrroleSupported, Nano Zerovalent Iron Nanocomposites for Adsorption of Hexavalent Chromium from Wastewater, **Chemistry Select.** 8 (2023) 1–13 (**IF: 2.30**).
- V. A. Waghmare, **R. Rathore**, A. Pandey, and V. Chandra, Highly efficient adsorption of aqueous iodine on polythiophene/ α -manganese dioxide nanocomposites, **Colloids Surfaces C Environ. Asp.** vol. 1, (2023) 100017.
- VI. A. Waghmare, **R. Rathore**, A. Pandey, V. Chandra, Highly Efficient Photocatalytic Degradation Efficiency of Polyaniline@Iron Phosphate Nanocomposite for Methyl Orange Dye under Solar Light, **Material Science and Engineering: B.** 302 (2024) 117253 (**IF: 3.89**).
- VII. **Roshni Rathore**, Aakash Waghmare, Sarita Rai, Vimlesh Chandra. "Synthesis and Characterization of MnO₂, MnFe₂O₄, and MnO₂@MnFe₂O₄ nanocomposite for Adsorption of MG dye from water." (Under communication).

- VIII. **Roshni Rathore**, Aakash Waghmare, Sarita Rai, Vimlesh Chandra. "Synthesis and Characterization of Mn_3O_4 nanoparticles for MG Dye Degradation under Solar Light." (Under communication).
- IX. A. Waghmare, **R. Rathore**, A. Pandey, V. Chandra, Polypyrrole based nanocomposite for wastewater treatment, Book Chapter. (Manuscript under preparation invited in Springer International Publishing).

9. CONFERENCE ACTIVITY

- I. Attended the 8th International Conference on Innovative Research Practices in Pharma, Science Management, Agriculture and Technology (IC-IRPPSMAT-2020) organized by Swami Vivekanand University Sagar M.P. held from Feb. 27th – 28th 2020.
- II. Presented e-poster titled "**Synthesis of MnO_2 nanoparticles for wastewater treatment**" at an online International seminar on Current Trends in Chemical and Pharmaceutical Sciences (ISCTCPS-2021) organized by Dr. Harisingh Gour Central University, Sagar M.P. from 2021st January 2021.
- III. Presented a paper titled "**Synthesis of $MnFe_2O_4$ Nanoparticles for Dye Removal**" at the International Conference on Energy and Environmental Materials (INCEEM-2021) online conference organized by the Department of Environmental Sciences and Physics, Sharda University, from 29th to 31st July 2021.
- IV. Presented a paper titled "Synthesis and Characterisation of Flower-like MnO_2 Nanocomposite with β -Dicalcium Silicates to Enhance Photocatalytic Activity" in the National Seminar on Applied Research (NSAR-2021), held online at Research Development Cell Sharda University, Greater Noida, UP, India from on 21st-22nd October 2021.
- V. A poster presentation titled "**Synthesis and Characterisation of $MnFe_2O_4$ Nanoparticles for Phosphate Removal**" in the International Symposium on Materials of the Millennium: Emerging Trends and Future Prospects (MMETFP-2021), held at the School of Technology, Pandit Deendayal Energy University, Gujarat from 19th-21st November 2021.
- VI. Presented a poster titled "**Wastewater treatment**" at the 10th International Conference on Drift and Research Shift in Pharm-Ag, Engineering, and Science (DRSPAES-2022) on 27-28 Feb. 2022 organized by Swami Vivekanand University, Sagar M.P.
- VII. Presented a paper titled (online mode) "Synthesis of α -Manganese Dioxide, β -Dicalcium Silicate, and α -Manganese Dioxide/ β -Dicalcium Silicate Nanocomposites for Effective **Removal of Phosphate from Water**" at the International Conference on Multidiscipline Academic Research and Innovation (ICMARI-2022) organized by Aryabhata Institute of Academics and Research, Lucknow on September 10-11, 2022.
- VIII. Presented a poster titled "**Photocatalytic Degradation of Crystal Violet Dye under Solar Light**" at the National Seminar on Recent Advance in Chemical Sciences (RACS-2023), organized by Dr. Harisingh Gour Central University Sagar, M.P. on March 16-17, 2023.

10. COMPUTER SKILLS:

Chem-draw, MS Word, PowerPoint, Excel, Origin, etc.

Roshni Rathore

Sagar (M.P.)